

CQF Exam Three.

Supervised Learning in Return Expectations

January 2021 Cohort

Short-term asset return is a challenging quantity to predict. Efficient markets produce near-Normal daily returns with no significant correlation between r_t, r_{t-1} . We do not expect supervised learning predictions to be accurate or robust out-of-sample – particularly for the simple set of features. This is an exercise.

- **GROUP A** asset can be a stock, ETF, market index or commodity – one ticker. FX not recommended but you can attempt. Predict direction only for a very short-term return (daily, or 4 hours, 15 mins). Group A is binomial classification: dependent variable is best labelled as 0, 1 vs. 1, -1.
Near-zero moves will be present in daily returns. If we bundle them with negative, that is likely to give an over-prediction in negative returns, therefore your choice to be based in your data.
- **GROUP B** time series is an index, factor, interest rate, or economic quantity for which you will be predicting 1M returns or changes (the longer-term) – one ticker or quantity. Given that the logistic classifier is formulated for binomial/multinomial prediction, it is recommended to explore a histogram and set up multinomial scheme, eg to predict buckets not necessarily a numerical value.

Feature	Formula	Description
O-C, H-L	Open - Close, High - Low	of price
Sign	$\text{sign} [r_t = \ln \frac{P_t}{P_{t-1}}]$	sign of return, sign of momentum
Past Returns	$r_{t-1}, r_{t-2}, \dots$	shift column of $t - 1$ to obtain $t - 2$
Momentum	$P_t - P_{t-k}$	price change period k days
Moving Average	$\text{SMA}_i = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}$	simple moving average
Exponential MA	$\text{EMA}_t = \text{EMA}_{t-1} + \alpha [P_t - \text{EMA}_{t-1}]$	recursive, $\alpha = 2/(N_{\text{obs}} + 1)$

Table 1: Features to choose from. Prediction either for t or $t + 1$, as appropriate.

- Volatility estimates, volume-driven or momentum-driven RSI, Stochastic K, MACD, CCI, Acc/Distrib, Bollinger Bands – while it may give an advantage to use TA indicators, you be creating a more complex assignment than necessary. We aim to learn the method, not guaranteeing a prediction.
- Do not use features from one asset to predict another. Without proper cointegration analysis, you are likely to simply introduce collinearity.
- Tutor will not reconfirm your choice of features. They will be of different length for Group B vs A and depend on your model.

Section 1 Classifier and Hyperparameters

1. Briefly present the maths of logistic classifier, use β for coefficients not θ , by writing down:
 - (a) the complete Loss Function (cost function) and relate it to MLE total log-likelihood function;
 - (b) the penalty expressions for Lasso and Ridge regressions (mathematical and graphical);
 - (c) RSS and MSE expressions, specific to logistic regression.
2. Implement Ridge and Lasso logistic regressions using the suitable full set of features in each prediction task Group A, Group B (a different set for each).
 - (a) produce a **table** comparing L1 and L2 type of penalisation: the impact made on regression coefficients;
 - (b) clearly explain whether L1 or L2 regression likely to have a high bias and low variance.

Section 2 Model Selection

1. Return to preprocessing and fit one model using 2 types of scaling on features data. Produce histograms of changed data columns. Report on changes to coefficients.
 - (a) min-max scaling or robust quantiles-based scaling;
 - (b) scaling to uniform $[0, 1]$ range.
2. Choose a restricted set of features, use penalisation and other (below), and produce a model that makes predictions with reduced variance (MSE). Accuracy score is **not** estimator variance.
 - (a) formally explain your approach to feature selection or elimination. For example, VIF for elimination of interdependent (colinear) features, or selection according to a score from the F-test.
 - (b) if prediction task allows (Group A, restricted model) graphically plot a logistic sigmoid for each feature. If prediction is not binomial (Group B), sigmoid only possible between two classes.
3. For the winning restricted model produce a suitable set of Evaluation metrics, such as area under ROC curve (each class) and confusion matrix. Give expressions for precision/recall.

Section 3 Mathematical bases of supervised learning

Question(s) in this section require mathematical working only, and not related to classifier implementation. The tutor can't provide further hints.

1. Consider the mean squared error $\text{MSE}(\hat{\beta})$ of estimator $\hat{\beta}$ in estimating the true value β

$$\mathbb{E} \left[(\hat{\beta} - \beta)^2 \right] = \text{Var}[\hat{\beta}] + \left(\mathbb{E}[\hat{\beta}] - \beta \right)^2.$$

Please answer below with Yes/No and one sentence of explanation.

- (a) can there exist an estimator with the smaller MSE than minimal least squares?
- (b) for a prediction, does the MSE measure an irreducible error or model error?

Instructions

Respond to all questions. E3 designed as one assignment: omitted questions might lead to a mark below passing. Requested mathematical and all computational workings (code) must be provided. Can create separate Python notebooks for **one Group A** and **one Group B**, however submit ONE PDF report and ONE ZIP file, name to start with LASTNAME.. Report should be analytical, with a. headings which match Sections/questions within and b. avoiding large output tables (do not print data).